the membrane there was contracting with precisely the speed of the ball's motion (speed of light). Just 0.001 second before reaching the critical circumference, the trapped ants dispatched ball number 14.999 (shown only in the last diagram). This ball, barely outracing the contracting membrane, did not reach the astronomer ant until 137 seconds after the collapse began. Ball number 15.001, sent out 0.001 second after the critical circumference, got inexorably sucked into the highly curved region and was crushed along with the five trapped ants.

But the astronomer ant could never learn about the crushing. She would never receive signal ball number 15, or any signal balls emitted after it; and those just before 15 would take so long to escape that to her the collapse would appear to slow and freeze right at the critical circumference.

This analogy is remarkably faithful in reproducing the behavior of an imploding star:

1. The shape of the membrane is precisely that of the curved space round the star—as embodied in an embedding diagram.
2. The motions of the signal balls on the membrane are precisely the same as the motions of photons of light in the imploding star's curved space. In particular, the signal balls move with the speed of light as measured locally by any ant at rest on the membrane; yet balls emitted just before number 15 take a very long time to escape, so long that to the astronomer ant the collapse seems to freeze. Similarly, photons emitted from the star's surface move with the speed of light as measured locally by anyone; yet the photons emitted just before the star shrinks inside its critical circumference (its horizon) take a very long time to escape, so long that to external observers the implosion must appear to freeze.
3. The trapped ants do not see any freezing whatsoever at the critical circumference. They are sucked through the critical circumference without hesitation, and crushed. Similarly, anyone on the surface of an imploding star will not see the implosion freeze. He will experience implosion with no hesitation, and get crushed by tidal gravity (Chapter 13).

This, translated into embedding diagrams, was the insight that came from Finkelstein's new reference frame. With this way of thinking about the